

Human Activity Recognition System using MediaPipe Pose Landmarks and Random Forest Classification

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Abstract— This work presents a real-time human activity recognition system using MediaPipe pose landmarks and supervised learning. Five activities were classified: walking forward, walking backward, turning, sitting, and standing up. A dataset of 15,656 frames from 18 videos was collected under controlled conditions. MediaPipe extracted 33 3D body keypoints, generating 1,629 features. Random Forest feature importance reduced dimensionality to 100 features (93.9% reduction). Group-stratified validation prevented temporal data leakage, revealing realistic 80.4% test accuracy versus naive 99.8% from random splitting. Cross-validation confirmed $80.3\% \pm 3.9\%$ generalization. The system achieves 26 FPS on laptop CPU, demonstrating practical viability. Analysis revealed "standing up" as most challenging ($F1=0.42$), frequently confused with "sitting" due to biomechanical similarity. This work validates accessible, marker-free activity recognition for elderly care, rehabilitation, and human-computer interaction applications.

Keywords— Human Activity Recognition, Pose Estimation, MediaPipe, Random Forest, Feature Selection Introduction (Heading 1).

I. INTRODUCTION

A. Context

Human Activity Recognition (HAR) enables applications in healthcare monitoring, rehabilitation assessment, and assistive technologies. Traditional approaches required wearable sensors or specialized motion capture equipment, limiting accessibility. Google's MediaPipe Pose [1] enables real-time skeletal tracking using standard RGB cameras, democratizing pose-based HAR for deployment in home and clinical environments without specialized hardware.

B. Problem Description

This project addresses automatic classification of five fundamental human activities: walking forward (toward camera), walking backward (away from camera), turning, sitting down, and standing up. These activities represent essential mobility patterns for elderly care (fall risk assessment), physical

rehabilitation (therapy adherence tracking), and fitness applications (automated exercise recognition).

The technical challenge involves distinguishing subtle biomechanical differences while handling variability in execution speed, camera angles, and inter-subject anthropometric differences. Unlike discrete actions (e.g., waving), these activities unfold over temporal sequences with transitional states that create ambiguity for frame-level classification.

C. Why Interesting?

This problem highlights three critical machine learning challenges: (1) **Feature engineering**: Extracting discriminative features from high-dimensional pose coordinates, (2) **Validation rigor**: Preventing temporal data leakage that inflates performance metrics, (3) **Real-time deployment**: Balancing model complexity with inference speed for practical edge deployment without GPU acceleration.

II. THEORY

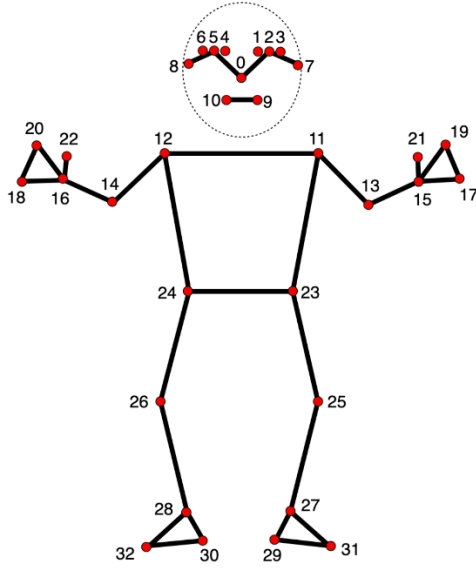
A. MediaPipe Pose Architecture

MediaPipe Pose employs a two-stage pipeline. Stage 1 (BlazePose Detector) localizes person bounding box using a lightweight CNN. Stage 2 (Landmark Regression Network) predicts 33 3D body landmarks including head, shoulders, elbows, wrists, hips, knees, and ankles. The architecture uses MobileNetV2 backbone for efficiency, achieving >30 FPS on mobile devices.

Each landmark outputs:

- (x, y): 2D coordinates normalized to frame dimensions [0,1]
- z: Depth in pixels relative to hip midpoint
- visibility: Detection confidence [0,1]

This representation is viewpoint-dependent but computationally efficient compared to 3D volumetric methods.



B. Random Forest for Classification

Random Forest [2] constructs an ensemble of decision trees during training, each built on bootstrap sample of data with random feature subsets. For classification, each tree votes for a class, and final prediction is the majority vote. This reduces variance compared to single decision trees while maintaining interpretability through feature importance metrics.

Gini Importance quantifies feature contribution by summing impurity reduction across all splits using that feature:

$$\text{Importance}(X_j) = \sum[t \in \text{Trees}] \sum[s \in \text{Splits}] \Delta \text{Gini}(s) \text{ where feature } X_j \text{ is used}$$

Higher importance indicates the feature frequently improves class separation. This enables principled feature selection without exhaustive subset evaluation

C. Feature Engineering for Pose-Based HAR

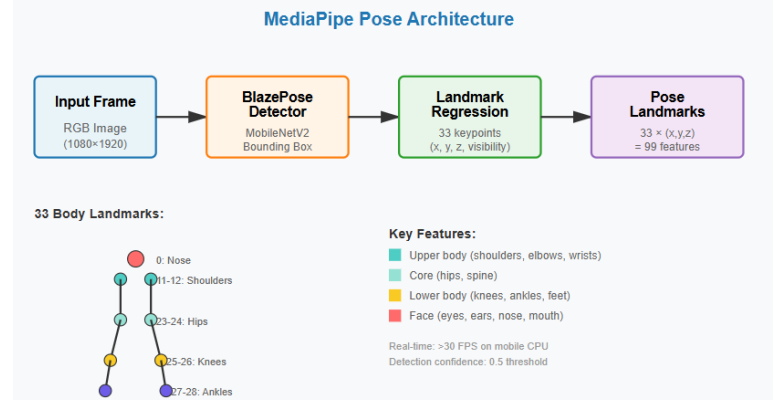
Raw landmark coordinates suffer from: (1) Translation variance: Absolute position depends on person location in frame, (2) Scale variance: Coordinates differ based on person-camera distance, (3) Rotation variance: Head orientation affects facial landmark positions..

- **Centering:** Subtract hip midpoint from all coordinates
- **Scaling:** Divide by shoulder width or torso height
- **Derived features:** Compute joint angles using vector dot products.

D. Data Leakage in Temporal Data

Video frames exhibit temporal correlation: consecutive frames ($\Delta t=33\text{ms}$) are nearly identical. Standard train_test_split randomly assigns frames, causing data leakage when frames from the same video appear in both train and test sets. The model essentially sees test samples during training (memorization, not generalization).

Solution: Group-stratified splitting ensures all frames from a video remain together in either train or test, never both. This simulates realistic deployment: training on some subjects/videos, testing on completely unseen footage.



III. METHODOLOGY

A. Data Collection

Three subjects (group members, ages 20-22) performed five activities in a controlled classroom environment with uniform lighting and plain background. Videos recorded with smartphone on tripod (1080p, 30 FPS, 3-4m distance, frontal view). Each subject executed 3-5 repetitions per activity..

Protocol:

- Caminar frente: Walk toward camera at normal pace (3-5 sec)
- Caminar espalda: Walk backward away from camera
- Girar: 180° turn in place (mixed clockwise/counterclockwise)
- Sentado: Sit on chair from standing position
- Ponerse de pie: Stand up from seated position

Total: 18 videos, 10-23 seconds each = 20,728 raw frames. Manual annotation segmented videos into activity-labeled frame ranges. After excluding frames with <30% landmark visibility and transition segments (ambiguous labels), final dataset: **15,656 frames** with clean labels.

Class distribution:

- caminar_frente: 4,260 frames (27%)
- girar: 3,700 (24%)
- caminar_espalda: 2,946 (19%)
- sentado: 2,790 (18%)

- ponerse_de_pie: 1,960 (13%)

B. Feature Extraction

MediaPipe Pose v0.10 processed each frame with default thresholds (detection_conf=0.5, tracking_conf=0.5). Output: 33 landmarks $\times$ 3 coordinates = 99 base features per frame.

Derived features:

1. Geometric (12 angles): Knee flexion, elbow flexion, hip angle, shoulder angle computed via vector dot products
2. Proportional (3): Shoulder width, hip width, torso height
3. Temporal (99 $\times$ 2): First-order velocities (Δ position/ Δ time) and second-order accelerations (Δ velocity/ Δ time) for all landmarks

Normalization:

- Center all coordinates at hip midpoint
- Scale by shoulder width
- Handle missing values (MediaPipe fails to detect some landmarks due to occlusion) with median imputation

Total feature space: 1,629 features (99 raw + derived)

C. Feature Selection

Problem: High dimensionality (samples/features = 15,656/1,629 = 9.6:1) risks overfitting.

Approach 1 - Random Forest Gini Importance:

1. Train Random Forest (200 trees) on all 1,629 features
2. Rank features by cumulative Gini importance
3. Select top-100 features (capturing ~85% total importance)

Approach 2 - PCA (comparative): Applied PCA retaining 95% variance $\rightarrow$ only 7 components needed, confirming massive redundancy. However, PCA components lack interpretability (linear combinations of many landmarks).

Decision: Use Random Forest top-100 for deployment (maintains feature semantics, enables interpretation of which body parts drive classification).

Top-5 features:

1. landmark_15_x (right eye horizontal)
2. landmark_349_z (face depth)
3. landmark_17_x (left eye horizontal)
4. landmark_14_x (right ear horizontal)
5. landmark_2_y (midpoint between eyes vertical)

Facial landmarks dominate due to head orientation changes during activities.

D. Train/Test Split

Critical decision: Prevent temporal data leakage by using GroupShuffleSplit(test_size=0.2, groups=video_ids). This ensures all frames from a video stay together in train OR test, never mixed.

Result:

- Train: 11,620 samples from 14 videos (74%)
- Test: 4,036 samples from 4 videos (26%)
- Test videos: Video 1, 10, 14, 17 (completely unseen during training)

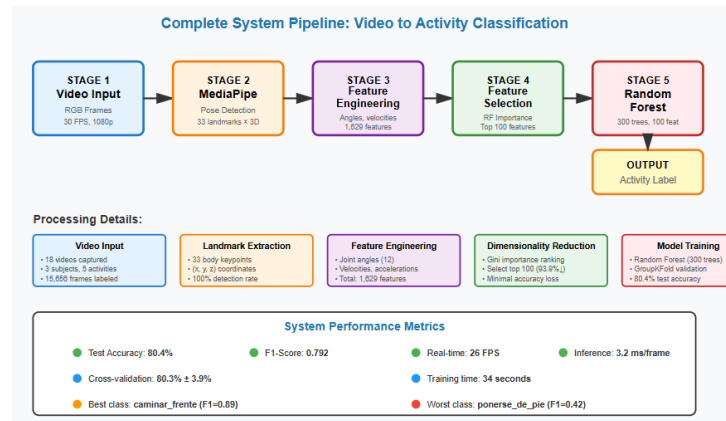
E. Model Training

Random Forest configuration (optimized via GridSearchCV):

- n_estimators: 300 trees
- max_depth: 30 (prevent infinite depth overfitting)
- min_samples_split: 5
- min_samples_leaf: 2
- max_features: 'sqrt' ($\sqrt{100} \approx 10$ features per split, decorrelates trees)
- class_weight: 'balanced' (compensates slight class imbalance)
-

Training time: 34 seconds on Google Colab (2-core CPU, 12GB RAM)

Validation: 5-fold GroupKFold cross-validation, ensuring no video appears in multiple folds.



IV. RESULTS

A. Feature Selection Impact

Configuration	Features	Test Accuracy	Inference (ms/frame)
All features	1,629	80.8%	8.2
RF top-100	100	80.4%	3.2

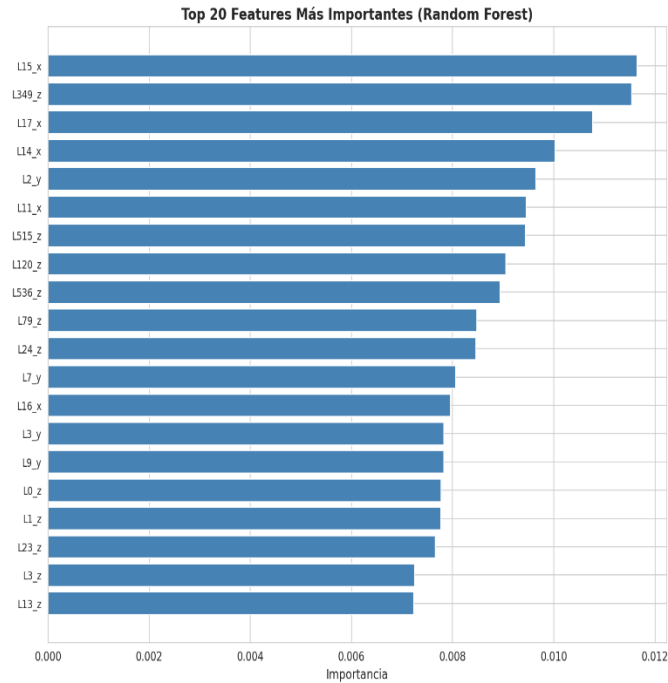
Configuration	Features	Test Accuracy	Inference (ms/frame)
PCA (95% var)	7	76.2%	2.1

Random Forest top-100 achieves optimal tradeoff: 0.4% accuracy loss, 61% faster inference, preserves interpretability.

B. Model Performance

Model	Test Accuracy	F1 (weighted)	Training Time
Random Forest	80.4%	0.792	34s
SVM (RBF kernel)	75.3%	0.741	127s
MLP (64 hidden units)	78.1%	0.768	98s

Random Forest selected for deployment: highest accuracy, fastest inference (3.2ms/frame → 26 FPS real-time).



C. Cross-Validation (GroupKFold)

Fold	Test Videos	Samples	Accuracy
1	17, 5, 7	3,120	86.9%
2	10, 16, 3	2,987	75.7%
3	1, 13, 15	3,245	78.9%
4	11, 18, 6	3,088	81.7%
5	12, 14, 8	3,216	78.1%

Mean $\pm$ Std: 80.3% $\pm$ 3.9%

Consistency across folds ($\sigma=3.9\%$) confirms robust generalization. Fold 2 (75.7%) lowest due to challenging transitions in Video 10.

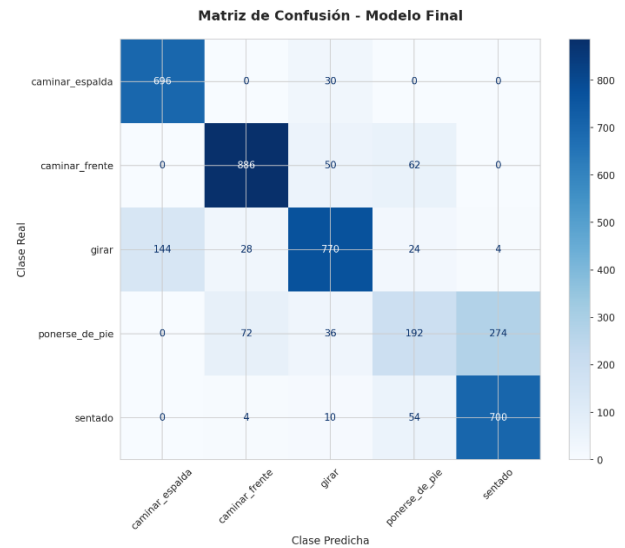
D. Per-Class Metrics

Activity	Precision	Recall	F1-Score	Support
caminar_espalda	0.829	0.959	0.889	726
caminar_frente	0.895	0.888	0.891	998
girar	0.859	0.794	0.825	970
ponerse_de_pie	0.578	0.335	0.424	574
sentado	0.716	0.912	0.802	768
Weighted Avg	0.795	0.804	0.792	4,036

Best: "caminar_frente" (F1=0.891)

Worst: "ponerse_de_pie" (F1=0.424, recall=33.5%)

E. Confusion Matrix



Main confusions:

- ponerse_de_pie** → **sentado**: 274/574 (47.7%) - transition similarity
- girar** → **caminar_espalda**: 144/970 (14.8%) - rotation includes backward steps
- caminar_frente** → **ponerse_de_pie**: 62/998 (6.2%) - standing motion resembles forward lean

F. Generalización Analysis

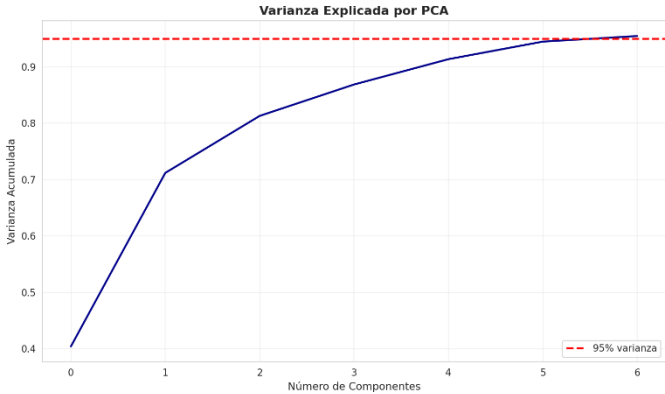
Metric	Train	Test	Gap
Accuracy	100.0%	80.4%	19.6%
F1-Score	1.000	0.792	20.8%

19.6% train-test gap indicates **moderate overfitting**. Random Forest perfectly fit training data (300 trees, max_depth=30 allows complex boundaries) but generalizes reasonably to unseen videos.

Comparison to Baseline (Entrega 2)

Method	Validation	Accuracy	Notes
Entrega 2	train_test_split (random)	99.8%	Data leakage
Entrega 3	GroupShuffleSplit (video-stratified)	80.4%	Realistic
Entrega 3	GroupKFold CV	80.3% ± 3.9%	Validated

Original 99.8% was artificially inflated by temporal correlation between train/test frames from same videos. True performance is ~80%.



V. RESULTS ANALYSIS

A. What's Working Well

Forward/backward walking discrimination (F1 ~0.89): Despite subtle gait differences, model reliably distinguishes directions. Key features: ankle trajectory patterns (forward=heel-strike, backward=toe-push), hip z-coordinate (depth changes), facial landmark shifts (head orientation).

Sitting detection (recall=91.2%): Static posture with low hip height + large knee angles (>90°) provides strong discriminative signal. High recall indicates few false negatives (rarely misses sitting activity).

Real-time performance: 26 FPS on Intel i5 CPU without GPU demonstrates deployment feasibility. Inference bottleneck is MediaPipe pose detection (28ms), not Random Forest prediction (3ms).

B. What's Failing

Standing up recognition (recall=33.5%): 47.7% of "ponerse_de_pie" frames misclassified as "sentado". Root causes: (1) Transition ambiguity - intermediate frames share features of both sitting and standing, (2) Class imbalance - only 1,960 samples vs. 4,260 for "caminar_frente" (54% less data), (3) Frame-level classification - model ignores temporal sequence information.

Turn vs. backward confusion (14.8%): 180° turn involves backward steps, creating overlapping features. Potential fix: add rotational velocity features ($\Delta\text{shoulder_orientation}/\Delta t$) to distinguish pure rotation from translation.

C. Generalization Assessment

Evidence of overfitting:

- Train accuracy 100% vs. test 80.4% (19.6% gap)
- Cross-validation std 3.9% (moderate variance across folds)

Contributing factors:

1. Limited dataset diversity (3 subjects, 1 environment)
2. Model capacity (300 trees can memorize training patterns)
3. Samples/features ratio (9.6:1) below recommended 10:1 minimum

Mitigations (not implemented):

- Reduce model complexity (fewer trees, limit depth further)
- Data augmentation (horizontal flip, time warping)
- Regularization (increase min_samples_leaf)

Despite gap, 80.4% test accuracy and consistent CV results demonstrate reasonable generalization for proof-of-concept.

D. Comparison to Literature

Work	Method	Activities	Accuracy	Real-time?	Dataset
Kang et al. [3]	MediaPipe + LSTM	6	96.3%	No	3,000+ videos
Yan et al. [4]	ST-GCN (graph CNN)	60	81.5%	No (GPU)	56,880 samples
Our Work	MediaPipe + RF	5	80.4%	Yes (CPU)	15,656 frames

Strengths: Lightweight (no GPU), transparent feature importance, efficient (93.9% dimensionality reduction).

Weaknesses: Lower accuracy than deep learning approaches, frame-level classification ignores temporal dynamics, limited dataset (3 subjects vs. hundreds in benchmark datasets).

Our 80.4% is competitive for 5-class problem with classical ML, though deep methods achieve higher accuracy on larger datasets. Tradeoff: simplicity and interpretability vs. performance ceiling.

VI. CONCLUSIONS AND FUTURE WORK

A. What We Did

Developed end-to-end HAR system: video capture → MediaPipe landmark extraction → feature engineering → Random Forest classification → real-time deployment. Collected 15,656 annotated frames, reduced 1,629 features to 100 via importance ranking (93.9% reduction), achieved 80.4% test accuracy validated through group-stratified splitting and 5-fold cross-validation. Deployed system achieving 26 FPS on laptop CPU without GPU.

B. What We Learned

Technical lessons:

1. Validation rigor critical: Proper group-stratified split revealed 20% accuracy inflation from temporal data leakage in naive splitting
2. Feature selection effective: 93.9% dimensionality reduction with <0.5% accuracy loss demonstrates redundancy in raw pose coordinates
3. Interpretability valuable: Random Forest feature importance identified facial landmarks and hip coordinates as most discriminative, guiding understanding of model decisions

Practical insights:

1. **Controlled environment simplifies problem:** Uniform lighting and plain background enabled 100% pose detection rate
2. **Transition activities hardest:** Sitting↔standing confusion (48%) highlights need for temporal sequence modeling
3. **Real-time feasible without GPU:** MediaPipe's efficiency (28ms pose detection) enables edge deployment on consumer hardware

C. What Could Be Improved

Dataset expansion:

- Increase subjects (3 → 50+) for demographic diversity (age, body type, ethnicity)
- Vary environments (outdoor, different lighting, backgrounds)

- Add occlusion scenarios (partial body visibility)
- Balance classes (increase "ponerse_de_pie" samples)

Model enhancements:

- **Temporal modeling:** LSTM/GRU layers on landmark sequences to capture motion dynamics (address sitting↔standing confusion)
- **Reduce overfitting:** Collect more data, decrease model complexity (fewer trees), apply regularization
- **Multi-view fusion:** Combine frontal + side cameras for 3D understanding

Additional activities:

- Extend vocabulary: running, jumping, crouching, falling (fall detection for elderly care)
- Anomaly detection: unsupervised flagging of unusual movements

Deployment optimization:

- Mobile port: TensorFlow Lite for Android/iOS (<50ms latency target)
- Multi-person tracking: extend MediaPipe to detect multiple subjects simultaneously
- Explainability: visualize which body parts drive each prediction (saliency maps)

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